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Palaeomagnetic studies of the Torridonian sediments, NW Scotland

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SYNOPSIS

This study covers thermal demagnetisation results from 172 localities distributed through the Upper Proterozoic Stoer Group and Torridon Group sediments of NW Scotland. Chemical leaching experiments and results from slump horizons confirm earlier studies showing that the bulk of the total NRM in these rocks resides in specularite grains and is related to deposition and/or initial lithification processes; in the Stoer Group and Aultbea Formation a considerable fraction of the remanence is held by magnetite. Thermal demagnetization partially, but not completely, removes that part of the small secondary remanence residing in the high blocking temperature fraction overlapping with the initial remanence. Demagnetization trends suggest that this secondary remanence is a composite one including both a Caledonian (*c.* 400 Ma) and a Tertiary/Recent component. The palaeopole estimates in stratigraphic order are: Stoer Group (main outcrop) 253°E, 46°N (A_{95} = 10°), Stoer Group (Enard Bay) 263°E, 37°N (A_{95} = 11°), Diabaig Formation 233°E, 17°S (A_{95} = 21½°), Cape Wrath sediments 237°E, 37°S (A_{95} = 16°), Lower Applecross Formation 230°E, 40°S (A_{95} = 16½°), Upper Applecross Formation 212°E, 27°S (A_{95} = 14½°), - Applecross Formation (Loch Maree), 231°E, 23°S (A_{95} = 17°), Aultbea Formation (Coigach Area) 231°E, 12°S (A_{95} = 8½°) and Aultbea Formation (main outcrop) 207°E, 7°S (A_{95} = 17°). The sampled sections complement and extend earlier studies to define in outline the magnetostratigraphy of these formations, and show that the contact between the Applecross and Aultbea Formations is strongly diachronous.

The time-sequence of poles from the Lower and Upper Torridonian coincide with the Laurentian a.p.w. path at *c.* 1100 and *c.* 1040 Ma respectively and imply that deposition and magnetization of these sediments predated isotopic closure by *c.* 130 Ma (Stoer Group, Rb-Sr isochron 968 Ma) and *c.* 260 Ma (Torridon Group, Rb-Sr isochron 777 Ma). This supports the views that the Torridonian and Moinian rocks are broadly coeval.

INTRODUCTION

The Hebridean Craton of NW Scotland is a peripheral part of the Laurentian Shield incorporating the bulk of Greenland and North America (Bridgwater *et al.* 1973). It includes the Torridonian sediments which have been of palaeomagnetic interest since early studies established that they possess a stable ancient magnetic

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remanence. The present study greatly extends the regional coverage of these sediments and examines the data in the wider context of Upper Proterozoic palaeomagnetism. Additional information is available as a Supplementary Publication (SUP 24015).

Geology and age

The Torridonian sediments are a thick, largely undeformed, suite of clastic sediments of predominantly fluviatile origin (Peach *et al.* 1907; Selley and Stewart 1967) lying unconformably on a degraded erosion surface of Lewisian gneiss. They include a major angular unconformity of 25–30° (Lawson 1965; Williams 1966; Stewart 1966) separating the Upper Torridonian from the Stoer Group (Stewart 1969), and are overlain by quartzites of Lower Cambrian age which dip at angles of up to 30° towards the Moine Thrust Zone of the Caledonian orogenic belt (*c.* 500–400 Ma) in the east. Final stages in the uplift of the older Lewisian complex are represented by K-Ar ages on biotites from Torridon of 1094 ± 42 Ma (Miller and Brown 1965), 1114 ± 30 Ma (Moorbath *et al.* 1967) and 1148 and 1169 Ma (Moorbath and Park 1972), and by a K-Ar whole-rock age from the Gairloch district of 1094 ± 45 Ma (Evans and Park 1965). Moorbath (1969) derived Rb-Sr isochrons of 968 ± 24 Ma and 788 ± 17 Ma from shale material near the middle of the Stoer Group and the base of the Applecross Formation respectively (all ages quoted here are recalculated according to constants listed by Steiger and Jäger 1977).

The Stoer Group (Lower Torridonian) is locally represented by up to 2 km of breccias overlain by fluviatile and lake sediments (Stewart 1969; Gracie and Stewart 1967) and contains the Stac Fada volcanic mud flow member. Our sample at Enard Bay (58°4′N, 5°20′W) includes the upper reworked part of the Stac Fada member (Fig. 1) and the Rudha Beag Sandstone unit. The Upper Torridonian ranges from 300 to 700 metres in thickness and was divided into three formations by Peach *et al.* (1907): a lower (Diabaig) comprising locally derived breccias passing laterally into grey shales of probable marine origin (Sutton 1962; Downie 1962), overlain by a middle (Applecross) and an upper (Aultbea) formation, the latter two reaching a maximum combined thickness of 5500 m. Study of detrital fragments in the Applecross Formation (Williams 1969) suggests a source area of supracrustal acid igneous and metamorphic rocks to the west, and deposition by a minimum of two coalescing fans, the lower from the NW and the higher from the SW; deposition took place on a subsiding bench on which the sea ultimately encroached and, as a result of overlapping of the fans, the facies boundaries are diachronous. K-Ar and Rb-Sr determinations on acid porphyry and metasedimentary material yield ages in the range 1800–1500 Ma (Moorbath *et al.* 1967) consistent with a derivation from terrains such as the Ketilidian mobile belt of south Greenland; an isotopic event at 1150–1100 Ma contributing to the detrital material corresponds with the late Gardar and Keweenawan events of the Laurentian Shield. The

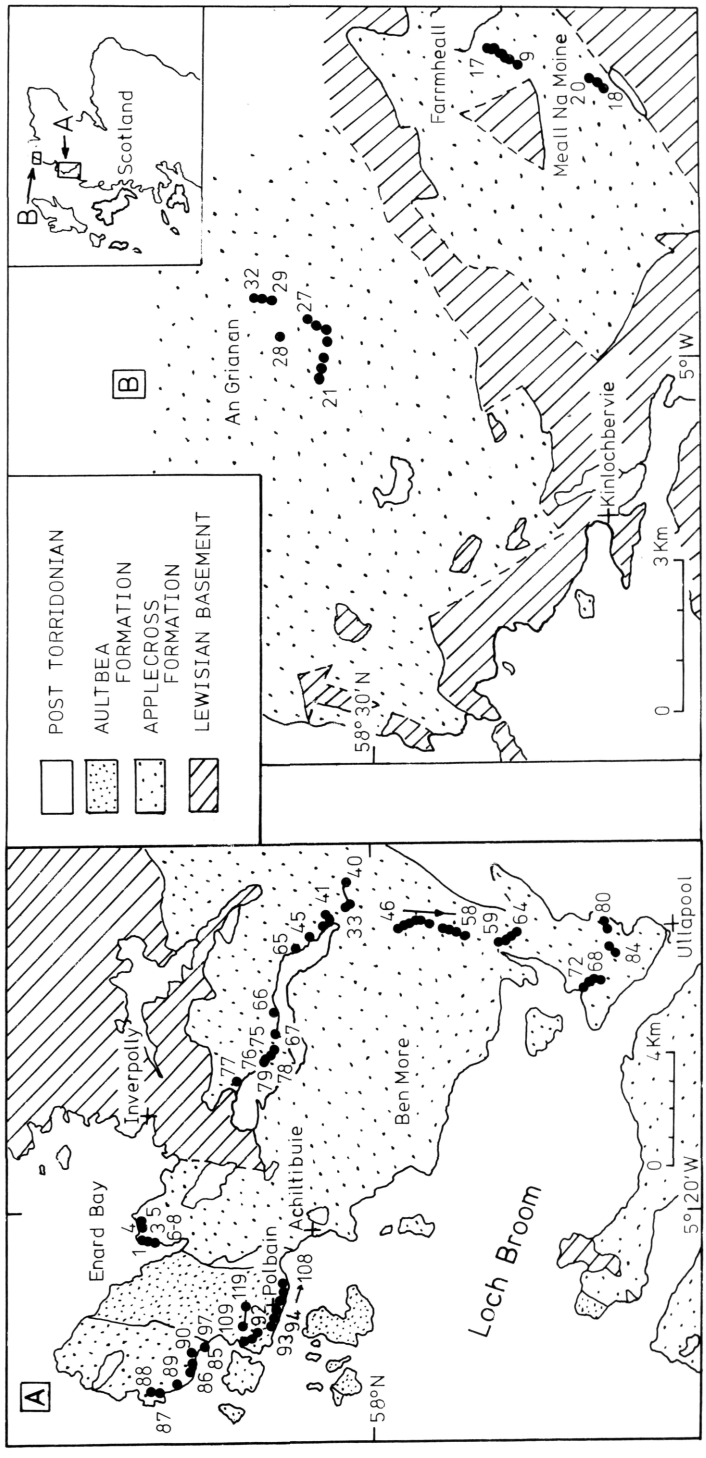


FIG. 1. Geology and palaeomagnetic sampling sites in the Torridonian sediments north of Loch Broom, NW Scotland.

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Torridonian outliers at Cape Wrath exhibit palaeocurrent directions comparable to the lower Applecross Formation in the Stoer region (Williams 1966).

Previous studies and objectives of the present work

The Torridonian sediments were included in one of the earliest systematic studies of this kind by Irving and Runcorn (Irving and Runcorn 1957; Irving 1957) who conducted comprehensive field tests to establish that the bulk of the characteristic magnetization was of depositional origin, carried by detrital specularite, with some rotation of the magnetic grains after deposition (Irving 1957); it is most closely described as a post-depositional DRM (Irving and Major 1964; Verosub 1977). The slump, fold and conglomerate tests suggest that the bulk of the magnetization predates lithification, but since the opposite polarities identified from the Upper Torridonian are not exactly antiparallel, it is possible that the total remanence is composite and includes one or more small secondary components. In a second study covering the Stoer Group and the base of the Torridon Group, Stewart and Irving (1974) were able to confirm that the change in remanence direction between the Lower and Upper Torridonian coincided exactly with the unconformity identified by Stewart (1966) and thermal cleaning to 500°C confirmed the stability of the remanence directions identified by Irving and Runcorn. The present study extends areal coverage of palaeomagnetic work within the Torridonian rocks and has the specific objectives of evaluating (i) the nature of the secondary magnetic remanence by stepwise thermal and chemical cleaning, (ii) the magnitude and direction of any apparent polar wander (a.p.w.) movements during the intervals of deposition of the lithological divisions, (iii) the magnetostratigraphic record, and (iv) the significance of the magnetisations and the Rb-Sr isochrons in the wider context of the a.p.w. history of the Laurentian Shield.

Sampling

Each palaeomagnetic site comprises an average of six separate cores drilled from the same bedded unit and oriented prior to extraction by sun sights and/or magnetic bearings. In stratigraphic sections the sampled sites are separated by thicknesses of between 5 and 50 m (Fig. 1); stratigraphic columns are derived from altitude determinations in conjunction with dip and strike measurements, comparisons with Geological Survey maps, and field studies of Peach *et al.* (1907), Williams (1966) and Stewart (1969, 1975). At the northernmost (Cape Wrath) outcrop stratigraphic sequences are sites 9–17 (Farrmheall), 18–20 (Meall na Moine) and 21–32 (An Grainan); regional locations north of Loch Broom are given in Fig. 1 and grid references in SUP 24015 Table I. Samples in the Loch Broom area cover (i) a stratigraphic section of some 2000 m through the Applecross Formation; (ii) the upper part of this formation (85–91, 104 m) and Aultbea Formation at

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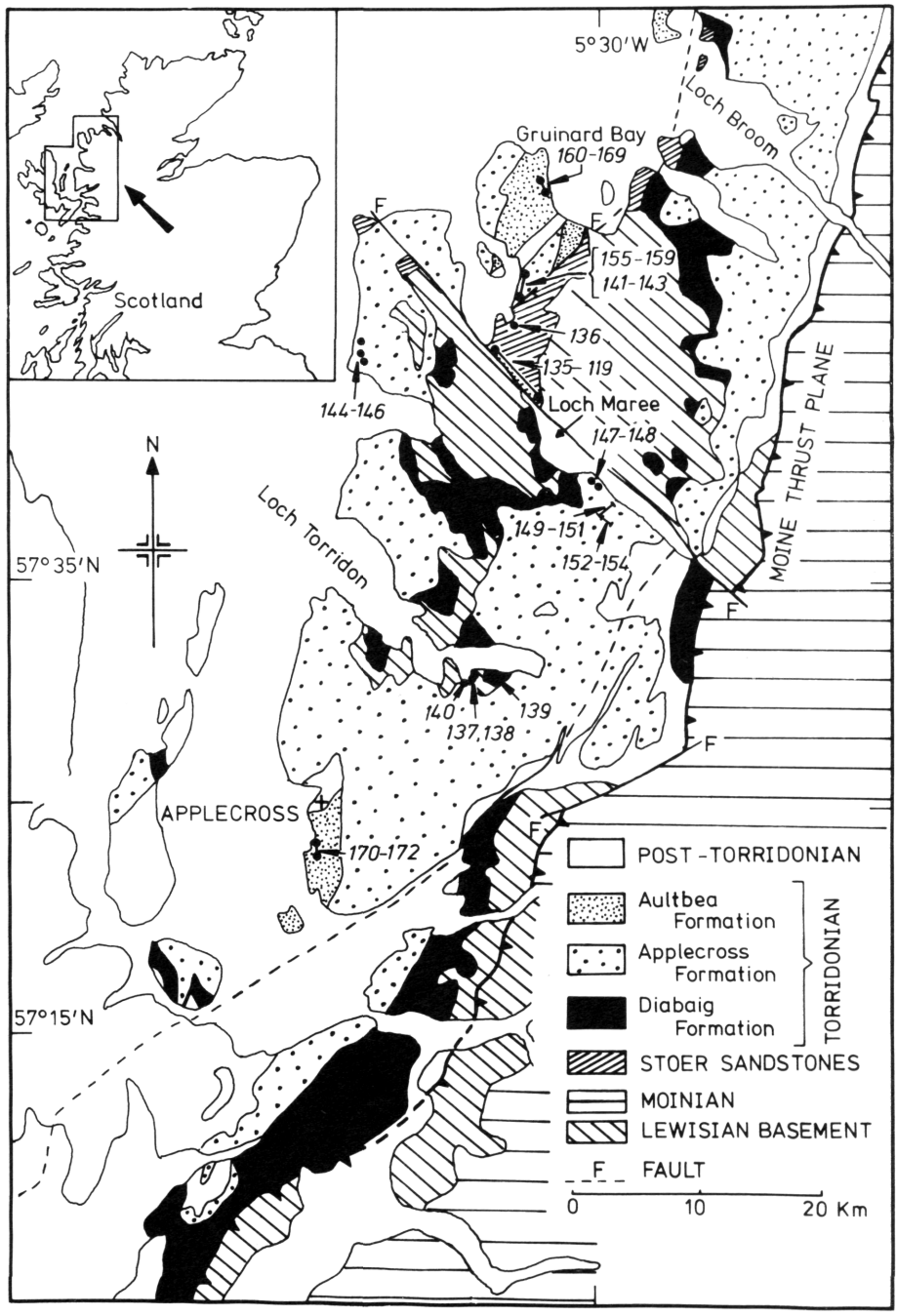


FIG. 2. Geology and palaeomagnetic sampling sites in the Torridonian sediments south of Loch Broom, NW Scotland.

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Rubha Mor (92–108, 114–119, two parallel sections covering *c.* 1200 m); (iii) beds at the base of the Applecross Formation at Loch Assynt (sites 73, 74). In the region south of Loch Broom (Fig. 2) 53 sites come from the Stoer Group (120–136) which is much thicker here than in the north, the Diabaig Formation (137–140), the Applecross Formation (141–159) and the Aultbea Formation (160–172).

The early studies established that the bulk of the remanence resides in detrital specular hematite, and polished section studies identify opaque minerals as angular grains of specular hematite, goethite as a replacement of specular hematite (widespread in the sandstones) and red ochre surrounding the specularite and staining silicate minerals (Stewart and Irving 1974). Although not yet detected optically, magnetite is suggested by our demagnetisation curves, notably from the Stoer Group, which show double blocking temperature spectra and double Curie points at *c.* 575°C (magnetite) and *c.* 675°C (hematite) (SUP 24015 Fig. 1). Magnetite has also been detected thermomagnetically by Parry (1957) and Schwarz (1969). These double Curie points are identified at a number of localities in the Stoer and Aultbea Formations (SUP 24015 Table 2), although the Applecross formation, in general, yields only single Curie points due to hematite.

Palaeomagnetic results

Summary of results

Alternating field demagnetization has, in general, little effect on the magnetizations of these samples (see also Stewart and Irving 1974) and thermal demagnetization was adopted as the routine method of treatment. For the large collections north of Loch Broom (work of J.E.F.S.) one or two pilot samples from each site were treated at intervals between 400° and 700°C in steps of 50°C reduced to 20°C above 550°C and the data were used to select one or two "blanket" temperatures for the treatment of the remaining samples. The treatment of pilot samples south of Loch Broom (work of R.L.S.) utilized steps of 100° to 500°C followed by 20°C steps above 580°C, and frequently identified different blocking temperature spectra at the same site; a range of cleaning temperatures was accordingly employed and calculated by combining the directions defining optimum groupings. As a result of this more detailed treatment, the within-site precisions are generally better in the second collection; all over-all means are calculated by the arbitrary exclusion of directions lying more than 40° from the group mean. All site mean directions are corrected for tectonic dips (mostly $< 15°$) as determined in the field or, where exposure was inadequate, estimated from the geological maps.

The results of a.f. and thermal demagnetization are described in the supplementary publication (SUP 24015). Frequently the cleaning treatment increases within-site dispersion but is a necessary complement to the study because it

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removes a secondary component and results in systematic movement of the remanence vectors (Fig. 3); this is believed to reflect the grain size effect observed by Irving (1957) where the internal dispersion increases with grain size which controls the degree of alignment with the palaeofield direction. Slump horizons in the Aultbea Formation yield random directions and fold tests yield an improvement in overall precisions but structural dips are not large enough to yield definitive tests. Chemical leaching experiments have successfully removed all of the red ochre pigment but show no significant reduction in the magnitude or direction of remanences in most samples, confirming that the bulk of this resides in the specularite grains; the exceptions are a few Applecross and Aultbea samples showing minor clockwise rotation of the vector similar to the results of thermal treatment.

Analysis of the minor secondary component of magnetization present in the Torridonian sediments is also described in the supplementary publication (SUP 24015); it identifies a component which is the resultant of at least two directions close to Caledonian and Tertiary/Recent palaeofield directions respectively. Because the blocking temperature spectra of this secondary remanence overlaps with the primary remanence, it cannot be entirely removed by cleaning treatment, but the overall group means are believed to be within 20° (and probably appreciably less) of the primary remanence direction. The collective data, together with the older data based on cleaned material (Stewart and Irving 1974) from the Stoer Group and basal Applecross Formation at Coigach, are summarized in Table 1.

Palaeomagnetic poles

The comprehensive study by Stewart and Irving (1974) through 600 m of Stoer Group sediments at Stoer Bay yields a direction of magnetisation (D = 313°, I = 29°) which is significantly less steep than that from some 1400 m of basal Stoer sediments of the main outcrop (Tournaig area) obtained in this study (D = 307°, I = 51°). The Stoer Group sediments thin northwards and at Stoer Bay the average palaeocurrent direction is towards 083°E compared with 321°E at Tournaig (Stewart 1975); this suggests that the Tournaig sediments are of a different age than those at Stoer Bay and implies that the mean 17° difference in pole positions between the two areas represents a.p.w. motion. If this is the case it requires that the time represented is fairly short: a mudflow correlated with the Stac Fada member occurs some 100 metres above site 136 with sites 132–135 several hundred metres below this being possibly not greatly different in age from sites 585 to 588 at Stoer (Stewart and Irving 1974); sites 120–132 are lower. A fairly rapid a.p.w. motion is consistent with interpretation of the collective data.

The four sites in the Diabaig sediments (pole 6, Table 1) yield positive SE directions in agreement with the study of Stewart and Irving (1974) which sampled the first 100 m of the formation in detail. No reversals were found although Irving

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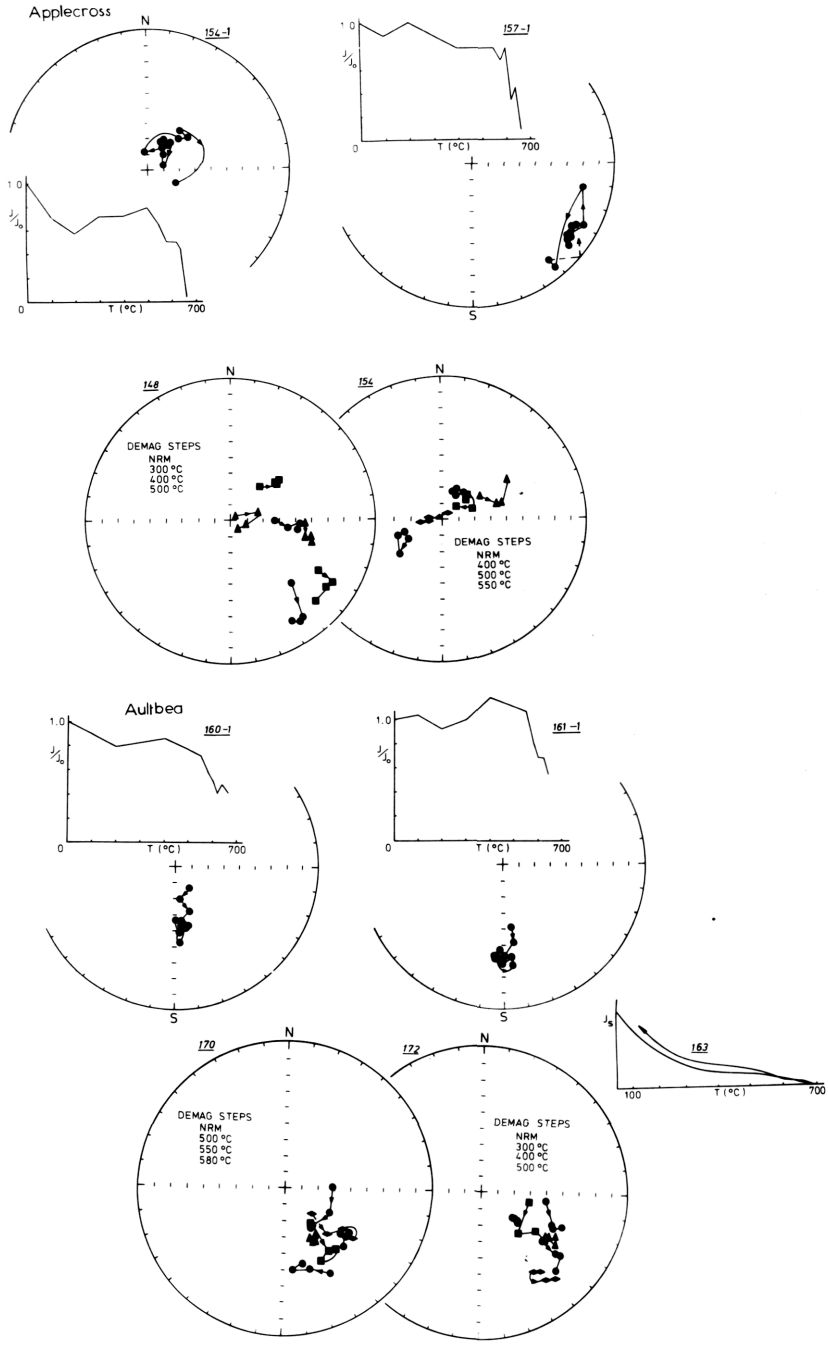


FIG. 3. Some representative palaeomagnetic results from the Applecross and Aultbea Formations of the Torridon Group. Stepwise thermal demagnetization of pilot samples is to 680°C or to a stage at which sample moments were no longer accurately measurable or direction behaviour was erratic. The collective results of blanket treatment at several temperatures of all samples from some sites is also shown. The thermomagnetic determinations show saturation remanence J_s as a function of temperature; the J_s axis is arbitrary.

TABLE 1

Summary of mean directions of magnetisation and pole positions derived from thermally-cleaned samples from divisons of the Torridonian rocks of NW Scotland

	Rock Division/Locality	D	I	N	R	k	α_{95}	Palaeomagnetic Pole °N	°E	(dp°, dm°)	Source
	LOWER TORRIDONIAN										
1.	Stoer Group Main Outcrop	307·0	51·5	13	12·5	23	8·8	46	252	(8, 12)	A
2.	Stoer Group Rubha Reidh	304·6	41·6	3	2·7	—	—	38	248	——	B
3.	Stoer Group Enard Bay	293·0	49·0	7	6·8	36·6	10·1	37	263	(9, 13)	A + B
4.	Stoer Group Achiltibuie	306·8	33·0	16	15·5	28·2	7·1	34	242	(5, 8)	B
5.	Stoer Group Stoer Bay	315·1	26·5	16	15·1	16·9	9·2	35	231	(5, 10)	B
	UPPER TORRIDONIAN										
6.	Diabaig Formation Diabaig	107·6	50·2	3	3·0	41	19·5	−17	233	(17, 26)	A
7.	Cape Wrath	91·3	62·9	8	7·7	24·3	11·5	−36	237	(14, 18)	A
8.	Applecross Formation, lower sites Coigach Area upper sites	92·7 124·2	68·2 67·0	17 18	15·7 16·8	12·1 13·7	10·7 9·7	−40·3 −27·0	229·8 211·7	(15, 18) (13, 16)	A
9.	Basal Applecross Formation Coigach Area	106	61	11	9·9	10	16	−28	228	(19, 24)	B
10.	Applecross Formation Loch Maree Area	106·5	56·4	10	9·3	13	13·7	−23	231	(14, 20)	A
11.	Aultbea Coigach Area	112·9	46·3	14	13·5	26·0	7·9	−12·2	231·3	(7, 10)	A
12.	Aultbea Main Outcrop	139·7	52·6	7	6·7	18	14·5	−7	207	(14, 20)	A

A = This study; B = Stewart and Irving (1974); N = number of sites used in computation of the mean, see Tables I and II (SUP 24015) and Stewart and Irving (1974). The original studies of these rocks by Irving (1957) yielded a palaeopole for the Stoer Group at 242°E, 35°N, and for the Upper Torridonian at 223°E, 6°S.

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and Runcorn identified a reversal near the top of this division. The palaeomagnetic poles for the Diabaig and Applecross Formations are the same within the limits of error. Stewart (1975) regards the Cape Wrath outlier (pole 7, Table 1) as the broad equivalent of the lowest part of the Applecross Formation and the similar pole positions support this. The palaeomagnetic pole for the Aultbea Formation in the type area (12, Table 1) is significantly different (at the 95% confidence level) from the Applecross poles and a.p.w. between times of deposition is inferred. The Aultbea Formation at Loch Broom (pole 11) yields a pole removed but not significantly so, from the Applecross Formation. Within the type locality there is no significant difference between the site mean directions from sites 170–172 at 150 m above the base of the formation and sites 160–169, 800 m higher up. As noted above, the evidence within the Upper Torridonian is that facies boundaries are diachronous, and while Irving and Runcorn (1957) found possibly as many as eight reversals of polarity in the type section of the Applecross, we do not detect reversals north of Loch Broom until the base of the Aultbea is reached; the Aultbea in this region may be the lateral equivalent of Applecross rocks further south. Pole 12 from the higher levels of the Aultbea Formation in the type area is regarded as the youngest palaeomagnetic pole of this study.

Magnetic stratigraphy

The collective palaeomagnetic studies (this study, Irving and Runcorn (1957) and Stewart and Irving (1974)) identify a number of reversals of polarity in the Applecross and Aultbea Formations, and these are combined to define the magnetic stratigraphy of the Torridonian rocks, in Figure 4. The correlations are made here purely on a lithological basis and these major lithological changes are unlikely to represent time planes. Indeed, this is a generalised statement of the magnetic stratigraphy in each area because the stratigraphic information required to place the sections in their correct relative positions is not available; we cannot at present know whether the reversals identified by Irving and Runcorn (1957) in different vertical sections in the southern part of the outcrop are to be correlated (a minimum of three NW-zones) or occur at later times towards the south as shown. Our sample from the Aultbea Formation south of Loch Broom derives from this unit as defined by the original Geological Survey work (Peach and Horne 1930) and these rocks are demonstrably above the "Applecross Formation" here. However, Stewart (1975) shows that the fine grained Aultbea *facies* occupies the whole of the Torridon Group section in this area, whereas north of Loch Broom there are 3 km of pebbly sandstones with the formal "Applecross" assignment below the fine grained Aultbea facies. Thus any correlation of time planes between the two areas could only be made with sections from the intervening ground. (We define the magnetisations as NW^- or SE^+ in the context of the directions of magnetisation because large a.p.w. movements in late Precambrian times (e.g.

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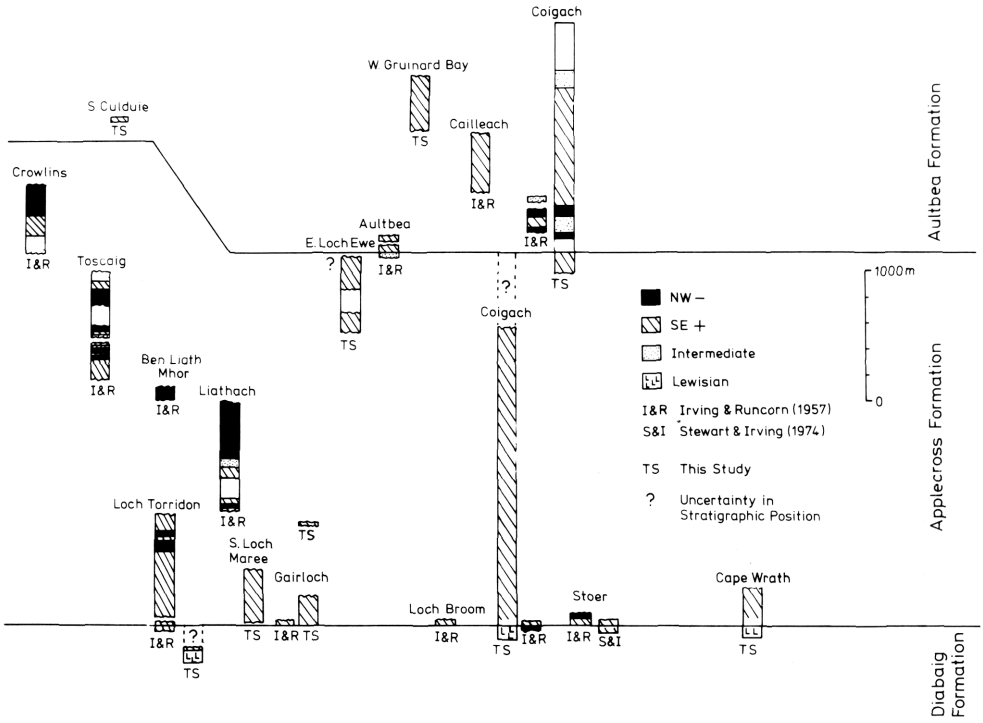


FIG. 4. A summary of the magnetic stratigraphy of the Stoer Group and Torridon Group rocks compiled from the results of Irving and Runcorn (1957), Stewart and Irving (1974) and this study. Because the normal or reversed nature of these Precambrian field directions is not yet fully resolved they are subdivided as NW− and SE+. Note the several reversals of magnetization in the thick development of the Applecross Formation in the south which are not present in the north, and the lithological divisions cannot represent time planes (see text). The data from this study are listed in Tables 1 and 2 (SUP 24015) with sampling sites in Figures 1 and 2. Irving and Runcorn (1957) give the comparable data in diagrammatic form for the sections in the south.

McElhinny *et al.* 1974; Morris and Roy 1977) leave some uncertainty about the normal or reversed nature of the polarities.)

DISCUSSION

Comparison with the Laurentian Upper Proterozoic a.p.w. path

The Torridonian sediments (Table 1) define two segments of the Upper Proterozoic a.p.w. path relative to the Hebridean region: pole 1 from the Stoer Group south of Tournaig is probably the oldest, and in the sporadic northern extension of the Group pole 3 includes basal members while poles 4 and 5 are more representative of higher members (Stewart and Irving 1974); the trend of poles

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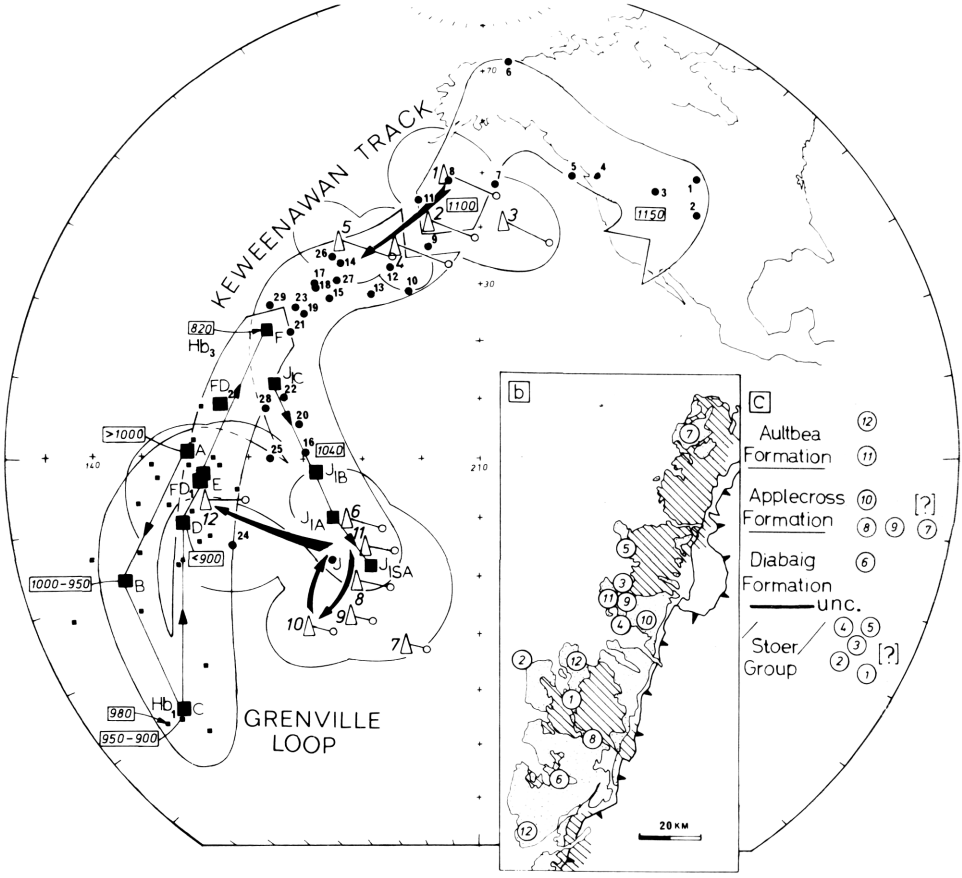


FIG. 5. A.p.w. path for the Laurentian Shield from *c.* 1150 to 820 Ma based mainly on McWilliams and Dunlop (1978), Roy and Robertson (1978), Piper (1977a) and Pesonen and Halls (1979). Palaeomagnetic poles from the Shield (small numbers) with assigned ages in brackets are: 1, Baraga County dykes (i); 2, Ilímaussaq Syenite (1144); 3, Lower Alona Bay lavas; 4, Baraga County dykes (ii); 5, Mamainse Lavas (R, 1070); 6, Lower Cape Gargantua Volcanics; 7, Marquette County dykes; 8, Oslerlavas (1115); 9, North Shore Volcanics (1110); 10, Duluth Complex (i), (1115); 11, North Shore Volcanics (1105); 12, Keweenawan intrusive (1120); 13, Beaver Bay Complex (1090); 14, Upper Gargantua Volcanics (1090); 15, Portage Lake lavas (1100); 16, Freda Sandstone; 17, Duluth Complex (ii); 18, Michipicoten Island lavas (952); 19, Copper Harbour lavas (1080); 20, Pikes Peak Granite (1040); 21, Isle Royale lavas; 22, Nonesuch shale (1060); 23, Basalt dykes, Minnesota; 24, Nankoweap Formation (1000); 25, Cardenas lavas (1090); 26, Logan dykes (1050); 27, Mamainse lavas (N, 1076); 28, Fond du lac Formation; 29, Kearsage rhyolite (1007). Poles J, and JIB-JISA are from the Jacobsville Formation, and mean uplift poles from the Grenville Mobile Belt (A–F) are calculated from thermochron zones defined by the K-Ar mineral ages indicated; individual pole determinations from the Province are indicated by small squares. Hb_1 and Hb_3 are dated Haliburton magnetisations, and FD_1 and FD_2 are mean poles from the Franklin diabase (*c.* 675–625 Ma). The Torridonian poles are plotted on the conventional reconstruction as small open circles and after the minor local rotation suggested in the

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plotting closest to the region (Fig. 5) is thus NE-SW although it is not unequivocally established that this represents a time sequence. The sequence of poles from the Upper Torridonian commences with the Diabaig Formation (pole 6, Table 1) followed by the basal Applecross in the Coigach area (pole 9), the bulk of the Applecross in this latter area (poles 8(a) and (b)) and the lower part of the Applecross in the Loch Maree area (pole 10); the small clockwise movement of the remanence vector suggested by data from this part of the Applecross (section 6(iv)) is an indication of a small loop in the a.p.w. path at this point. The relationship of the Aultbea pole (sections 6(v) and 6(xiii)) to the Applecross/Aultbea further south is uncertain but Aultbea pole 12 is certainly younger and defines the youngest part of the path in Figure 5.

The removal of these poles from the post-1000 Ma a.p.w. path for the Laurentian Shield (see Irving and McGlynn 1976, figs. 3 and 4) is important confirmation of the primary nature of the bulk of the remanence in these sediments. These two segments of a.p.w. path for NW Scotland form a basis for establishing the configuration of the Hebridean Craton in the Laurentian Shield, and are plotted in Figure 5 together with the Upper Proterozoic a.p.w. path for the Laurentian Shield between 1150 and 820 Ma. Poles older than 1150 Ma are excluded because this is the age of the youngest thermal events in the underlying Lewisian Complex (Moorbath and Park 1972). In addition polar data between 1180 and 1040 Ma in age define a rapid motion of the pole known as the "Great Logan Loop" (Robertson and Fahrig 1971) with the earlier part of this motion comprising a movement of the pole from the central Pacific region to the vicinity of the Shield between 1180 and 1160 Ma (Piper 1977a,b) and clearly unrelated to the Torridonian a.p.w. motions. The return path of the pole towards the central Pacific region by 1040 Ma is defined by a range of rocks related to the Keweenawan rifting and magmatism, and a recent assessment of these data is given by Pesonen and Halls (1979). In addition poles JIC to JISA are the sequence of probable primary remanence directions from the Jacobsville Formation of the Lake Superior region (Roy and Robertson 1978) broadly linked with the Keweenawan rifting but of uncertain age. The link between all these poles and the sequence of uplift magnetizations from the Grenville Province is not yet clearly defined and the latter are expressed as mean poles in Fig. 5 calculated from thermochron zones linked to uplift ages in the range 1050–900 Ma (A–D) and following McWilliams and Dunlop (1978). The thermochron zones are only rough indicators of the age of remanence but two more specific determinations of the ages of poles Hb_1 and Hb_2 at 980 and 820 Ma have been made by thermochronometry (Berger *et al.* 1979); individual poles from the Province are plotted as small squares in Figure 5.

text as open triangles with 95 percent confidence ovals. The location of the Torridonian studies in the Precambrian foreland of NW Scotland is shown in Fig. 5b where the Lewisian (metamorphic) outcrop is shaded and the Torridonian outcrop is stippled (Fig. 5c is a generalised stratigraphic succession).

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The palaeomagnetic data from the Torridonian rocks match the movements and position of the Laurentian path at 1100 Ma and 1040 Ma using the conventional North Atlantic continent fit of Bullard *et al.* (1965) (open circles in Fig. 5) based on the present continental margins. The match is improved however, if the Hebridean region is rotated clockwise by 8° ± 4° about a local axis (triangles in Fig. 5). This optimum fit is equivalent to a clockwise rotation of 38° about a Euler pole at 27.7°E, 88.5°N, and a local clockwise rotation of 8° about a Euler pole at 345°E, 57.5°N. There is no support for a 30° anticlockwise rotation (Beckmann 1979). The palaeomagnetic data would fit this correlation by moving the Hebridean region in the conventional reconstruction (Bullard *et al.* 1965 and Fig. 5) up to about 200 km to the NW but further adjustments are restricted by the criteria that (i) NW Scotland should not overlap the Laurentian Shield, (ii) it does not lie within the Grenville terrain since it is not overprinted by the *c.* 1100–800 Ma thermal events, and (iii) the Upper *and* Lower Torridonian data must collectively fit the a.p.w. path. These constraints preclude a closer match between the Torridonian data and post-1000 Ma.

The 8° discrepancy between the conventional and Upper Proterozoic reconstructions, if real, must predate Mesozoic rifting, and in the context of North Atlantic continental drift, the Hebridean region is exceptional because rifting did not follow Phanerozoic lineaments. Large scale late Caledonian movements along this belt at *c.* 400 Ma are in a sinistral sense (Morris 1976; Piper 1979) appropriate to the anti-clockwise rotation of a unit incorporating the Hebridean Craton.

Age of the Torridonian magnetizations and regional implications

The Upper and Lower Torridonian formations yield excellent Rb-Sr isochrons of 968 and 777 Ma (Moorbath 1969, recalculated to constants listed by Steiger and Jäger 1977) but their physical significance is uncertain (Berger *et al.* 1979). Correlation with the dated Laurentian poles of Figure 5 imply that the magnetization (and depositional) ages are *c.* 1100 Ma and 1040 Ma respectively and unrelated to these isochrons; it requires that the isotopic systems did not homogenize until 130 Ma and 260 Ma after deposition. Such discrepancies are possible because, while older Rb-Sr ages may result from inherited radiogenic strontium (Chaudhuri 1976), the younger ages result from post-depositional diagenetic reactions between the mineral phases; Perry and Turekian (1974) have shown that homogenization is not quite complete at 5500 m in Miocene shales and comparable findings are reported by Bath (1974). A syn- or immediate pre-Grenville age for the Torridonian is also consistent with the geological evidence: the rocks have previously been widely equated with the Moinian metasediments overlying Lewisian inliers of the Northern Highlands in the immediately-adjoining Caledonian orogenic belt. A Grenville-age magmatic/tectonic event at *c.* 1000 Ma is recognised in the SW sector of the Northern Highlands (e.g. Brook *et al.* 1977). Deposition of the Moinian

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and Torridonian, and earliest tectonic phase in the Moinian should probably be linked to the evolving mobile belt, and the *c.* 750 Ma Morarian event can currently only be linked with pegmatite intrusion into Moinian rocks. It appears from Figure 5 that the Torridonian rocks are coeval with the infilling of the higher parts of the Keweenawan rift, and the Upper Torridonian is specifically contemporaneous with the Jacobsville succession.

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References

BATH, A. H. 1974. New isotopic data on rocks from the Long Mynd, Shropshire. *J. Geol. Soc.* **130,** 567–74.

BEACH, A. 1975. The interrelations of fluid transport, deformation, geochemistry and heat flow in early Proterozoic shear zones in the Lewisian complex, *Philos. Trans. R. Soc. Lond.* **A280,** 569–604.

BECKMANN, G. E. J. 1979. A review of Nagssugtoqidian palaeomagnetism. *Rapp. Grønlands Geol. Unders.* **89,** 115–24.

BERGER, G. W., YORK, D. and DUNLOP, D. J. 1979. Calibration of Grenvillian palaeopoles by $^{40}Ar/^{39}Ar$ dating. *Nature* **227,** 46–8.

BRIDGWATER, D., WATSON, J. and WINDLEY, B. F. 1973. The Archaean craton of the North Atlantic region. *Philos. Trans. R. Soc. Lond.* **A 273,** 493–512.

BROOK, M., POWELL, D. and BREWER, M. S. 1977. Grenville events in the Moine rocks of the Northern Highlands, Scotland. *J. Geol. Soc.* **133,** 489–96.

BULLARD, E., EVERETT, J. E. and SMITH, A. G. 1965. The fit of the continents around the Atlantic. *Philos. Trans. R. Soc. Lond.* **A 258,** 45–51.

CHAUDHURI, S. 1976. The significance of the Rubidium-Strontium age of sedimentary rock. *Contrib. Mineral Petrol.* **59,** 161–70.

DOWNIE, C. 1962. So-called spores from the Torridonian. *Proc. Geol. Soc. Lond.* **1600,** 127–8.

DUNLOP, D. J. 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. *Geophys. J. R. Astron. Soc.* **27,** 37–55.

EVANS, C. R. and PARK, R. G. 1965. Potassium argon age determinations from the Lewisian of Gairloch, Ross-shire, Scotland. *Nature* **205,** 350–2.

GRACIE, A. J. and STEWART, A. D. 1967. Torridonain sediments at Enard Bay, Ross-shire *Scott. J. Geol.* **3,** 181–94.

HARLAND, W. B. 1980. Comment and reply on "A palaeomagnetic pole position from the folded Upper Devonian Catskill red beds and its tectonic implications." *Geology* **8,** 258–9.

IRVING, E. 1957. The origin of the palaeomagnetism of the Torridonian sandstone series of north-west Scotland. *Philos. Trans. R. Soc. Lond.* **A 250,** 100–10.

—— and MAJOR, A. 1964. Post-depositional detritial remanent magnetisation in a synthetic sediment. *Sedimentology,* **3,** 135–43.

—— and McGLYNN, J. C. 1976. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia. *Philos. Trans. R. Soc. Lond.* **A 280,** 433–68.

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 20, 2015

—— and RUNCORN, S. K. 1957. Analysis of the palaeomagnetism of the Torridonian sandstone series of north-west Scotland. *Philos. Trans. R. Soc. Lond.* **A 250,** 83–99.

LAWSON, D. E. 1965. Lithofacies and correlation within the Lower Torridonian. *Nature* **207,** 706–8.

McELHINNY, M. W. 1964. Statistical significance of the fold test in palaeomagnetism. *Geophys. J. R. Astron. Soc.* **8,** 338–40.

—— GIDDINGS, J. W. and EMBLETON, B. J. J. 1974. Palaeomagnetic results and late Precambrian glaciations. *Nature Phys. Sci.* **248,** 557–61.

McWILLIAMS, M. O. and DUNLOP, D. J. 1978. Grenville palaeomagnetism and tectonics. *Can. J. Earth. Sci.* **15,** 687–95.

MILLER, J. A. and BROWN, P. E. 1965. Potassium-argon age studies in Scotland. *Geol. Mag.* **102,** 106–34.

MOORBATH, S. 1969. Evidence for the age of deposition of the Torridonian sediments of north-west Scotland. *Scott. J. Geol.* **5,** 154–70.

—— and PARK, R. G. 1972. The Lewisian chronology of the southern region of the Scottish mainland. *Scott. J. Geol.* **8,** 51–74.

—— STEWART A. D., LAWSON, A. E. and WILLIAMS, G. E. 1967. Geochronological studies on the Torridonian sediments of north-west Scotland. *Scott. J. Geol.* **3,** 389–412.

MORRIS, W. A. 1976. Transcurrent motion determined palaeomagnetically in the Northern Appalachians and Caledonides and the Acadian orogeny. *Can. J. Earth Sci.* **13,** 1236–43.

—— and ROY, J. L. 1977. Discovery of the Hadrynian Polar Track and further study of the Grenville problem. *Nature* **266,** 689–92.

PALMER, H. C. 1970. Palaeomagnetism and correlation of some Middle Keweenawan rocks, Lake Superior. *Can. J. Earth Sci.* **7,** 1410–36.

—— HALLS, H. C. and PESONEN, L. J. 1976. The great conglomerate test. *Phys. Can.* **32,** 37.

PARRY, J. H. 1957. The problem of reversed magnetisations and its study by magnetic methods. *Adv. Phys.* **6,** 299.

PEACH, B. N. and HORNE, J. 1930. *Chapters on the Geology of Scotland,* Oxford University Press.

—— HORNE, J., GUNN, W., CLOUGH, C. T., HINXMAN, L. W. and TEALL, J. J. H. 1907. The Geological Structure of the North-West Highlands of Scotland. *Mem. Geol. Surv.*

PERRY, E. A. and TUREKIAN, K. K. 1974. The effects of diagenesis on the redistribution of strontium isotopes in shales. *Geochim. Cosmochim. Acta* **38,** 929–35.

PESONEN, L. J. and HALLS, H. C. 1979. The palaeomagnetism of Keweenawan dikes from Baraga and Marquette Counties, North Michigan. *Can. J. Earth Sci.* **16,** 2136–49.

PIPER, J. D. A. 1977a. Magnetic stratigraphy and magnetic-petrologic properties of Precambrian Gardar lavas, South Greenland. *Earth Planet. Sci. Lett.* **34,** 247–63.

—— 1977b. Palaeomagnetism of the Giant dykes of Tugtutoq and Narassoq Gabbro, Gardar Igneous Province, South Greenland, *Bull. Geol. Soc. Denmark* **26,** 85–94.

—— 1979. Aspects of Caledonian palaeomagnetism and their tectonic implications. *Earth Planet. Sci. Lett.* **44,** 176–92.

PULLAIAH, G., IRVING, E., BUCHAN, K. L. and DUNLOP, D. J. 1975. Magnetisation changes caused by burial and uplift. *Earth Planet. Sci. Lett.* **28,** 133–43.

ROBERTSON, W. A. and FAHRIG, W. A. 1971. The Great Logan Palaeomagnetic Loop—the polar wandering path from Canadian Shield rocks during Neohelikian Era. *Can. J. Earth Sci.* **10,** 1279–301.

ROY, J. L. and ROBERTSON, W. A. 1978. Palaeomagnetism of the Jacobsville Formation and the apparent polar path for the interval −1100 to −670 m.y. for North America. *J. Geophys. Res.* **83,** 1289–304.

SCHWARZ, E. J. 1969. Magnetochemical aspects of heating of red and green beds. *Pap. Geol. Surv. Can.* **70–63,** 31.

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 20, 2015

SELLEY, R. C. and STEWART, A. D. 1967. The Torridonian sediments of NW Scotland. *7th Int. Sediment. Congr. Excursion Guide C2.*

SMITH, R. L. and PIPER, J. D. A. 1982. Palaeomagnetism of the Southern Zone of the Lewisian (Precambrian) Foreland, NW Scotland. *Geophys. J. R. Astron. Soc.* **68,** 325–47.

STEIGER, R. H. and JAGER, E. 1977. Subcommission on Geochronology: Convention on the use of decay constants in Geo- and Cosmochronology. *Earth Planet. Sci. Lett.* **36,** 359–62.

STEWART, A. D. 1966. An unconformity in the Torridonian. *Geol. Mag.* **103,** 462–4.

—— 1969. Torridonian rocks of Scotland reviewed, *In* M. Kay (ed), *North Atlantic Geology and Continental Drift. Mem. Am. Assoc. Pet. Geol.* **12,** 595–608.

—— 1975. Torridonian rocks of western Scotland, *In* Downie, C., Harland, W. B. and Moorbath, S. (eds), *Precambrian, Geol. Soc. Lond. Spec. Rep.* No 6, 43–51.

—— and IRVING, E. 1974. Palaeomagnetism of Precambrian sedimentary rocks from NW Scotland and the apparent polar wandering path of Laurentia. *Geophys. J. R. Astron. Soc.* **37,** 51–72.

SUP 24015. Material deposited with, and available from, the British Library (Lending Division), Boston Spa, Wetherby; West Yorkshire LS23 7BQ, UK.

SUTTON, J. 1962. Torridonian microfossils. *Geol. Mag.* **99,** 379.

VEROSUB, K. L. 1977. Depositional and post-depositional processes in the magnetisation of sediments. *Rev. Geophys. Space Phys.* **15,** 129–43.

WILLIAMS, G. E. 1966. Palaeography of the Torridon Applecross Group. *Nature* **209,** 1303–6.

—— 1969. Petrography and origin of pebbles from Torridonian strata (late Precambrian), northwest Scotland. *In* Kay, M., (ed), *North Atlantic Geology and Continental Drift, Mem. Am. Assoc. Pet. Geol.* **12,** 609–29.

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